MODEL SELECTION & IMPROVEMENT

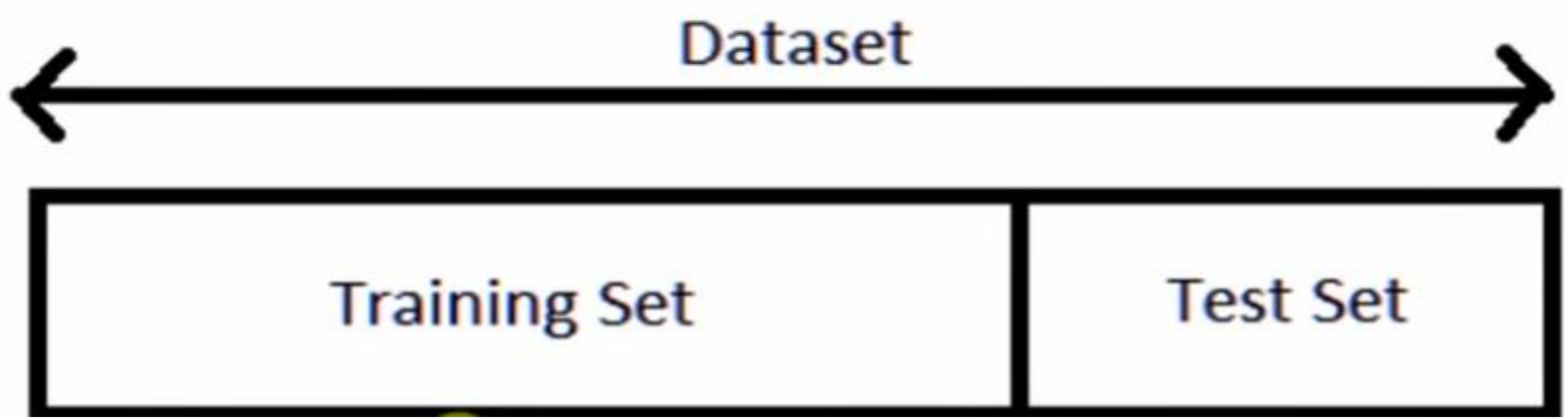
- **Cross-validation is a statistical technique used in machine learning and statistics to assess the performance and generalizability of a predictive model.**
- **The primary purpose of cross-validation is to provide a more reliable estimate of a model's performance on an independent dataset than a single train-test split.**

Cross-validation is like testing different models to find the best one in machine learning. It ensures models work well with various data parts and helps make them adaptable. Think of it as finding the right outfit for different occasions.

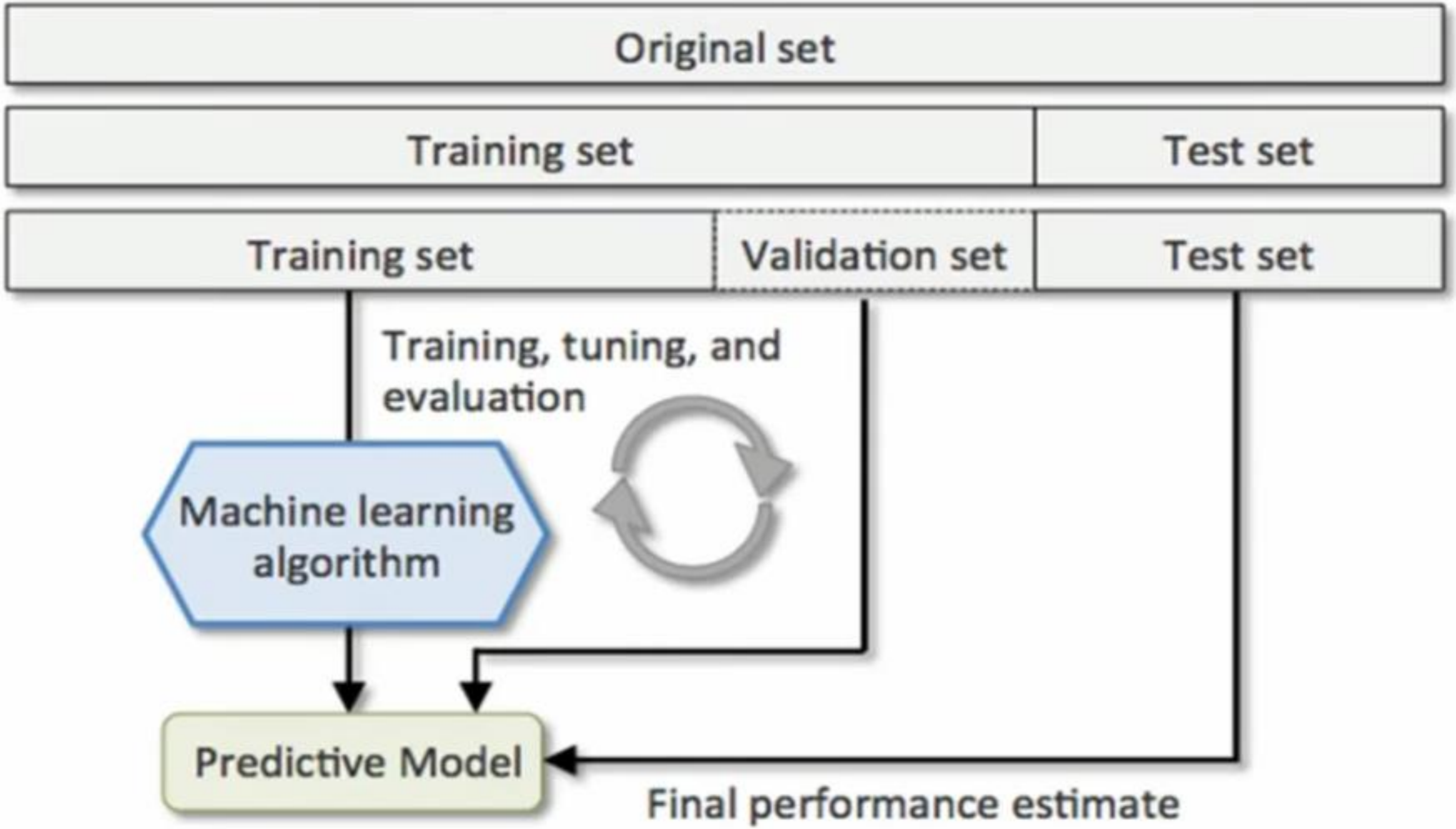
Here are some key reasons why cross-validation is used:

i. Model Evaluation
ii. Reducing Variance
iii. Maximizing Data Utilization
iv. Detecting Overfitting
v. Hyperparameter Tuning
vi. Model Selection
vii. Imbalanced Datasets
viii. Real-world Performance Estimation

Train Test datasets in Machine Learning



Train Test Validation datasets in Machine Learning



Cross Validation in Machine Learning

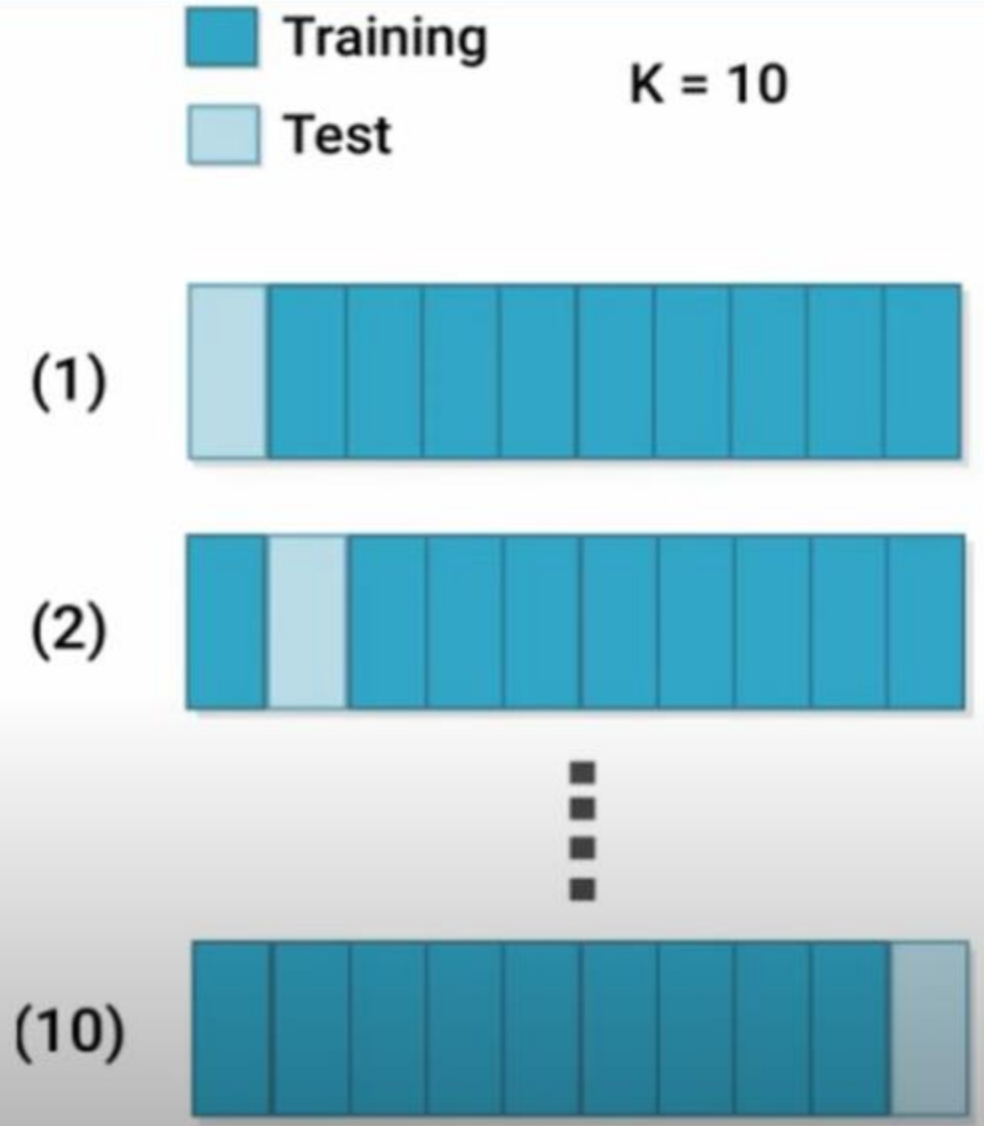
- The idea of cross-validation arises because of the *problems* with train test model or train test validation model.
- It basically wants to guarantee that the score of our model does not depend on the way we picked the train and test set.

Types Cross Validation in Machine Learning

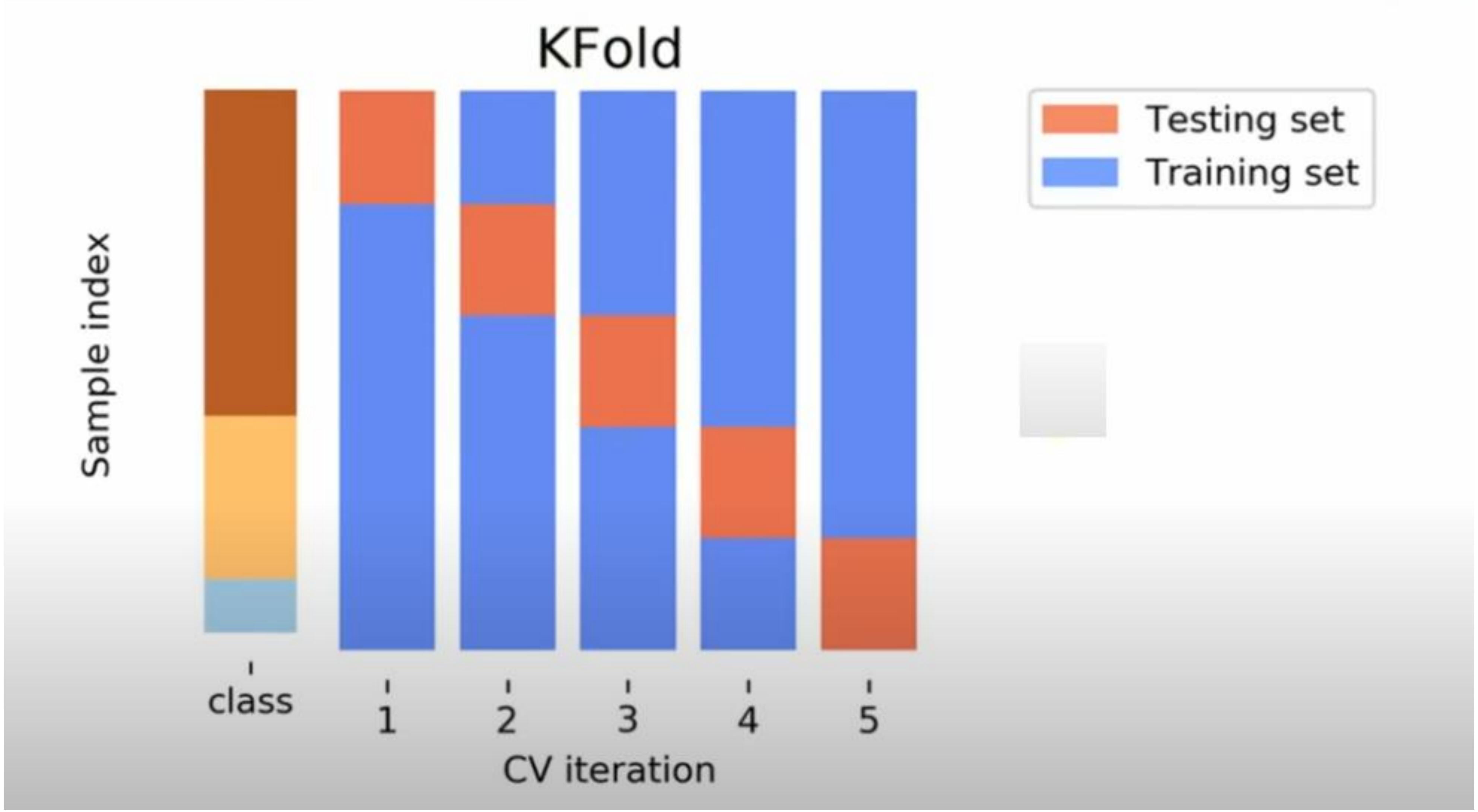
Following are the type of Cross Validation Techniques

- K-folds
- Stratified K-folds
- Leave-one-out
- Leave-p-out

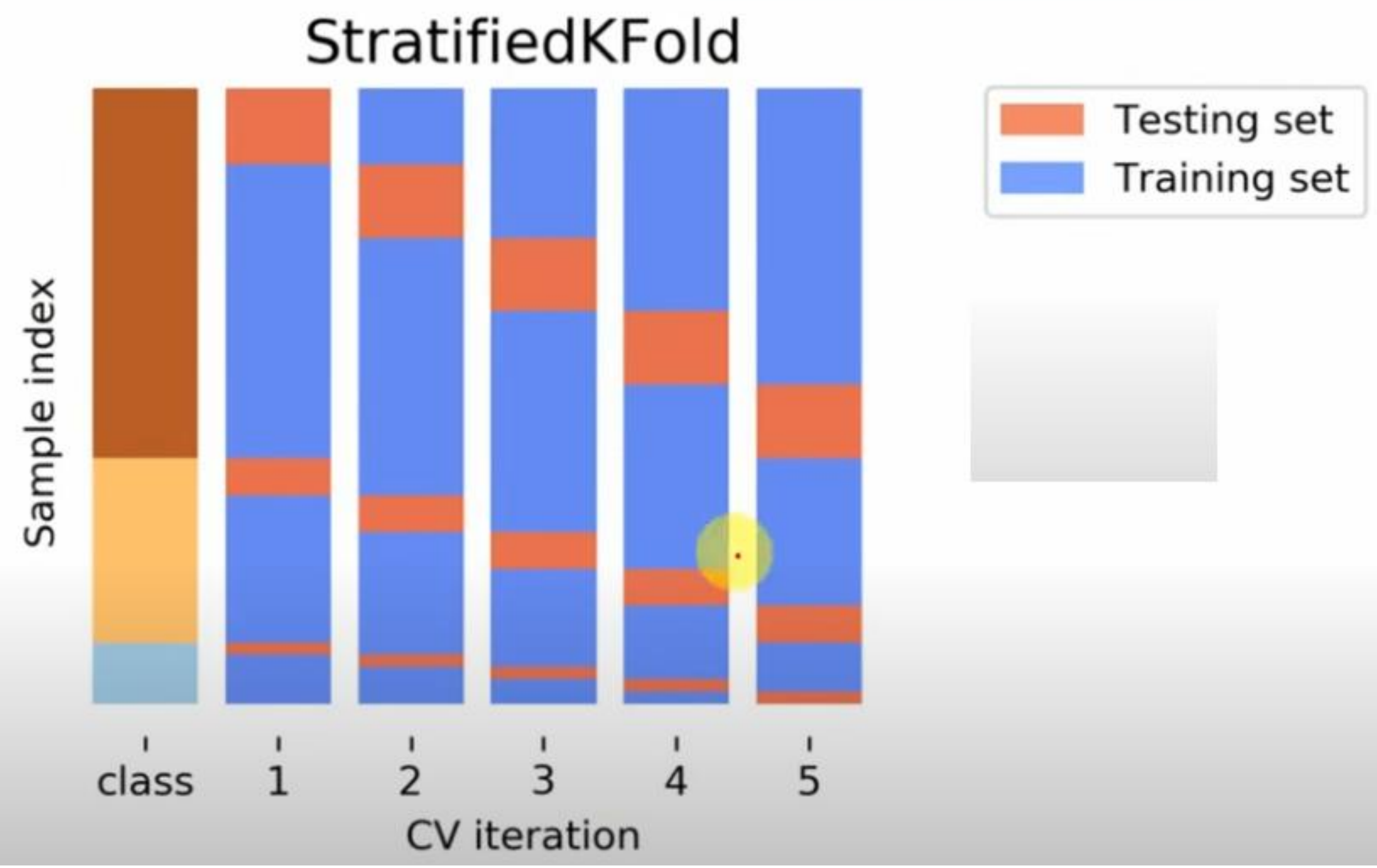
K-Fold Cross Validation in Machine Learning



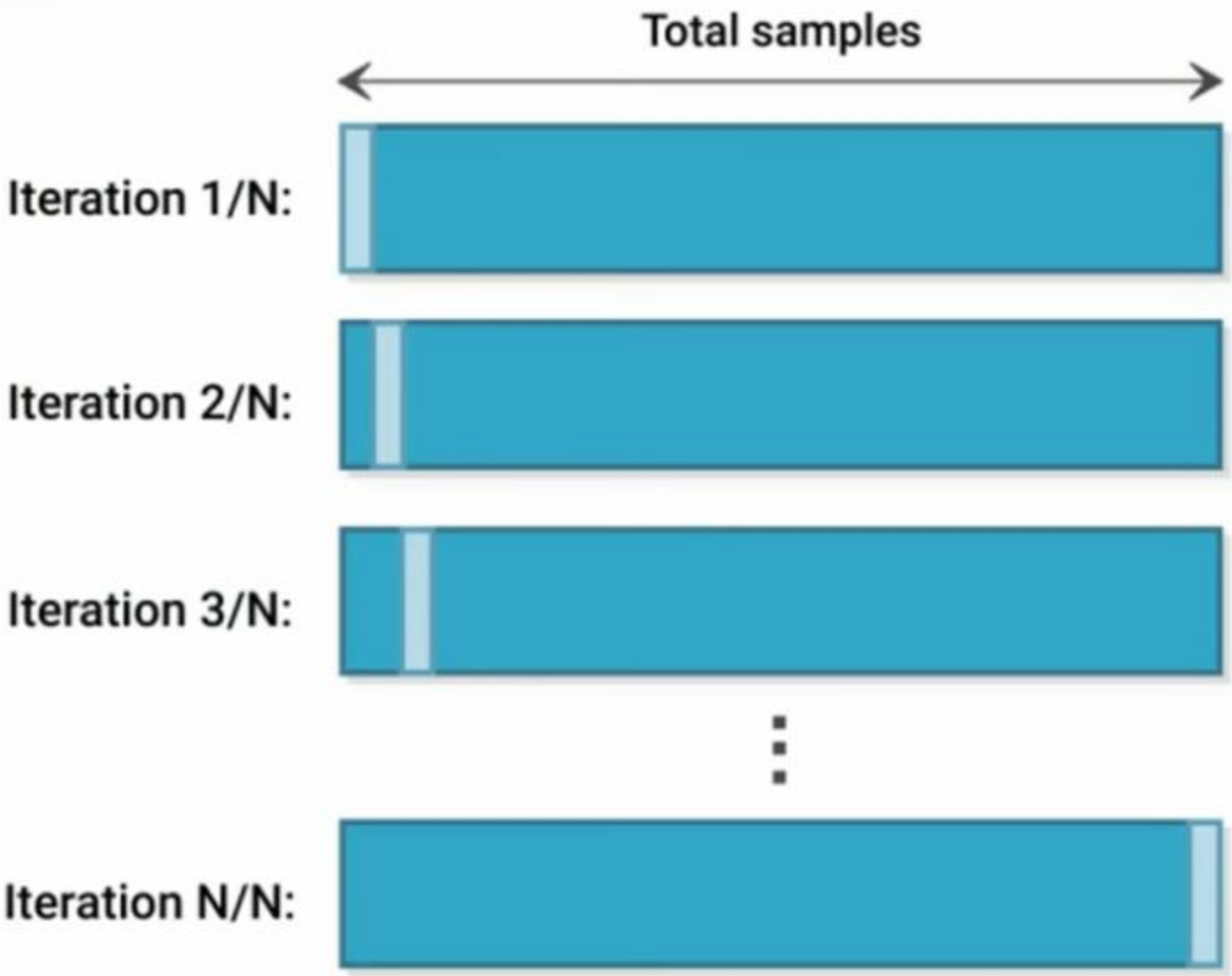
K-folds Cross Validation in Machine Learning



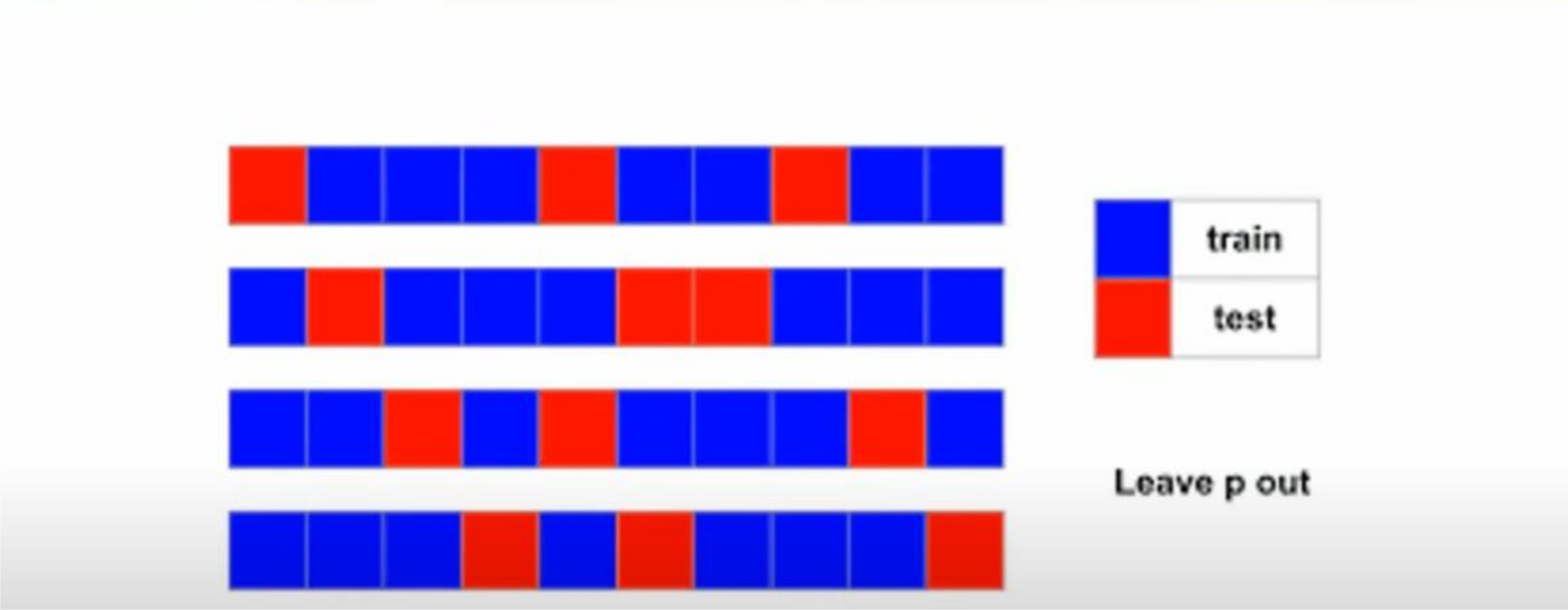
Stratified K-folds Cross Validation in Machine Learning



Leave-one-out Cross Validation in Machine Learning



Leave-P-out Cross Validation in Machine Learning



Dimensionality Reduction

Dimensionality Reduction

- The number of input features, variables on columns present in a given dataset is known as dimensionality and the process to reduce these features is called Dimensionality Reduction.
- It is a way of converting the higher dimensions dataset into lesser dimensions data ensuring that it provides similar information.

Uses:

It is used in:

- Speech Recognition
- Signal Processing
- Bioinformatics
- Data Visualization
- Noise Reduction

Dimensionality Reduction Types

<u>Feature Extraction</u>:

- In feature extraction, we are interested in finding a new set of k features that are the combination of the original n features.
- These methods may be supervised or unsupervised depending on whether or not they use the output information.
- The best known and most widely used feature extraction methods are Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA).

Dimensionality Reduction Types

<u>Feature Selection</u>:

- In feature selection, we are interested in finding k of the total of n features that give us the most information and we discard the other (n-k) dimensions.
- The best known and most widely used feature selection methods are Filter methods, Wrapper methods and Wrapper methods.